On the largest prime factor of integers in short intervals IV

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1 Introduction

The famous Legendre's conjecture states that there is always a prime number between consecutive squares. Let x be a sufficiently large integer, then it is conjectured that there is always a prime number in the interval $[x, x+x^{\frac{1}{2}}]$. In 1969, Ramachandra [13] showed that this interval contains a number with a prime factor larger than $x^{0.576}$. The exponent 0.576 has been improved by many authors, and the following table lists some improved exponents.

Ramachandra [14]	0.625
Graham [4]	0.662
Zhu [16]	0.675225
Jia [6]	0.692
Baker [1]	0.7
Jia [7]	0.71
Jia [8], Liu [11]	0.723
Jia [9]	0.728
Baker and Harman [2]	0.732
Liu and Wu [12]	0.738
Harman [5]	0.74
Baker and Harman [3]	0.7428
Li [10]	0.7437

Table 1

In this short note, by inserting new exponent pairs into the estimates for exponential sums, we improve the exponent to 0.744.

Theorem 1.1. *For sufficiently large x, the interval $[x, x+x^{\frac{1}{2}}]$ contains an integer with a prime factor larger than $x^{0.744}$.*

The exponent 0.744 can be slightly improved by more accurate numerical calculations.

2 Proof of Theorem 1.1

Throughout this note, we always suppose that ε is sufficiently small and $0.5 < \theta \leqslant 0.75$. The letter p is reserved for prime numbers. Let

$$\mathcal{A} = \{n : x^\theta < n \leqslant ex^\theta,\ N(n) = 1\}, \quad \mathcal{B} = \{n : x^\theta < n \leqslant ex^\theta\},$$

$$\mathcal{A}_d = \{a : ad \in \mathcal{A}\}, \quad P(z) = \prod_{p<z} p, \quad S(\mathcal{A}, z) = \sum_{\substack{a \in \mathcal{A} \\ (a,P(z))=1}} 1$$

and write $S(\theta) = S\left(\mathcal{A}, \left(ex^\theta\right)^{\frac{1}{2}}\right)$ that counts the number of primes in $\mathcal{A}$. By the discussions in [2], we only need to show that

$$\int_{0.6-\varepsilon}^{0.744} \theta S(\theta) d\theta < (0.4 - \varepsilon) \frac{x^{\frac{1}{2}}}{\log x}. \tag{1}$$

Next, we use [[12], Theorem 3(3.2)] with $(\kappa, \lambda) = (k_n, l_n)$ $(-50 \leqslant n \leqslant -2)$, where (k_n, l_n) are exponent pairs in [[15], Table 5.2]. After verifying all the conditions needed, we can prove an analog of [[5], Lemma 6.3] that holds for $0.6 \leqslant \theta \leqslant 0.746$, with ϕ_j and $\tau(\theta)$ are replaced by the values in Table 2.

Now, by our new $\tau(\theta)$ together with the discussions in [[10], Subsection 4.6], we can show that

$$\int_{\frac{5}{8}-\varepsilon}^{\frac{2}{3}-\varepsilon} \theta S(\theta) d\theta < 0.0946 \frac{x^{\frac{1}{2}}}{\log x}. \tag{2}$$

Note that in [10] we got 0.0975. For $\frac{2}{3} - \varepsilon \leqslant \theta \leqslant 0.744$, we can use the following upper bound for $S(\theta)$ given in [[2], Lemma 17] with a small modification:

$$\begin{aligned} S(\theta) \leqslant \Bigg(& \frac{2}{\tau(\theta) - \theta + 0.9} - \int_{\theta-\frac{1}{2}}^{\tau(\theta)} \frac{\omega\left(\frac{\theta-t_1}{t_1}\right)}{t_1^2} dt_1 - \int_{\frac{\tau(\theta)-\theta}{3}+0.3}^{\theta-\frac{1}{2}} \int_{\theta-\frac{1}{2}}^{\tau(\theta)} \frac{\omega\left(\frac{\theta-t_1-t_2}{t_2}\right)}{t_1 t_2^2} dt_2 dt_1 \\ & - \int_{\frac{\tau(\theta)-\theta}{3}+0.3}^{\theta-\frac{1}{2}} \int_{t_1}^{\theta-\frac{1}{2}} \int_{\theta-\frac{1}{2}}^{\tau(\theta)} \frac{\omega\left(\frac{\theta-t_1-t_2-t_3}{t_3}\right)}{t_1 t_2 t_3^2} dt_3 dt_2 dt_1 \Bigg) \frac{x^{\frac{1}{2}}}{\log x}, \end{aligned} \tag{3}$$

where $\omega(u)$ denote the Buchstab function. Using (3), we have

$$\int_{\frac{2}{3}-\varepsilon}^{0.744} \theta S(\theta) d\theta < 0.268 \frac{x^{\frac{1}{2}}}{\log x}. \tag{4}$$

From [10], we know that

$$\int_{0.6-\varepsilon}^{\frac{5}{8}-\varepsilon} \theta S(\theta) d\theta < 0.0352 \frac{x^{\frac{1}{2}}}{\log x}. \tag{5}$$

By (2), (4) and (5), we conclude that

$$\int_{0.6-\varepsilon}^{0.744} \theta S(\theta) d\theta < 0.398 \frac{x^{\frac{1}{2}}}{\log x}. \tag{6}$$

Clearly (1) holds from (6), and the proof of Theorem 1.1 is completed.

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Appendix: Values of ϕ_j and $\tau(\theta)$ $(\phi_j - \varepsilon \leqslant \theta < \phi_{j+1} - \varepsilon)$

j	1	2	3	4	5	6
ϕ_j	$\frac{3}{5}$	$\frac{11}{18}$	$\frac{579}{898}$	$\frac{10317}{16000}$	$\frac{1625}{2502}$	$\frac{3529}{5314}$
$\tau(\theta)$	$2-3\theta$	$\frac{1}{6}$	$\frac{449\theta-123}{999}$	$\frac{475583\theta-15108}{93433}$	$\frac{819\theta-267}{1567}$	$\frac{3177\theta-1077}{5845}$
j	7	8	9	10	11	12
ϕ_j	$\frac{85}{127}$	$\frac{1151551}{1713181}$	$\frac{14141}{20982}$	$\frac{645}{949}$	$\frac{181}{262}$	$\frac{313}{446}$
$\tau(\theta)$	$\frac{435419\theta-150809}{783240}$	$\frac{1317\theta-521}{2011}$	$\frac{566241\theta-236683}{795067}$	$\frac{14437\theta-6187}{19453}$	$\frac{274\theta-127}{320}$	$\frac{1063\theta-503}{1191}$
j	13	14	15	16	17	18
ϕ_j	$\frac{83}{117}$	$\frac{745}{1042}$	$\frac{1063}{1478}$	$\frac{487}{674}$	$\frac{974}{1343}$	$\frac{2533}{3482}$
$\tau(\theta)$	$\frac{1867\theta-895}{2037}$	$\frac{3050\theta-1475}{3268}$	$\frac{2357\theta-1147}{2493}$	$\frac{6973\theta-3409}{7305}$	$\frac{9953\theta-4883}{10351}$	$\frac{13792\theta-6785}{14262}$
j	19	20	21	22	23	24
ϕ_j	$\frac{1075}{1474}$	$\frac{4033}{5518}$	$\frac{2483}{3391}$	$\frac{2011}{2742}$	$\frac{7243}{9862}$	$\frac{8605}{11702}$
$\tau(\theta)$	$\frac{9320\theta-4595}{9594}$	$\frac{24659\theta-12179}{25291}$	$\frac{32023\theta-15839}{32745}$	$\frac{40918\theta-20263}{41736}$	$\frac{25771\theta-12775}{26231}$	$\frac{64105\theta-31805}{65133}$
j	25	26	27	28	29	30
ϕ_j	$\frac{1688}{2293}$	$\frac{11821}{16042}$	$\frac{13693}{18566}$	$\frac{5251}{7114}$	$\frac{9005}{12191}$	$\frac{20473}{27698}$
$\tau(\theta)$	$\frac{78829\theta-39139}{79971}$	$\frac{95948\theta-47669}{97210}$	$\frac{57854\theta-28759}{58548}$	$\frac{138367\theta-68815}{139887}$	$\frac{164195\theta-81695}{165853}$	$\frac{193474\theta-96299}{195276}$
j	31	32	33	34	35	36
ϕ_j	$\frac{7717}{10434}$	$\frac{26053}{35206}$	$\frac{14594}{19711}$	$\frac{10855}{14654}$	$\frac{36193}{48838}$	$\frac{40081}{54062}$
$\tau(\theta)$	$\frac{113249\theta-56387}{114225}$	$\frac{263573\theta-131273}{265681}$	$\frac{305017\theta-151955}{307287}$	$\frac{351160\theta-174985}{353598}$	$\frac{201172\theta-100267}{202478}$	$\frac{458923\theta-228779}{461715}$
j	37	38	39	40	41	42
ϕ_j	$\frac{7373}{9941}$	$\frac{48673}{65602}$	$\frac{53395}{71942}$	$\frac{19471}{26226}$	$\frac{31868}{42911}$	$\frac{69373}{93386}$
$\tau(\theta)$	$\frac{521263\theta-259903}{524241}$	$\frac{589742\theta-294095}{592912}$	$\frac{332375\theta-165775}{334059}$	$\frac{746689\theta-372469}{750261}$	$\frac{835973\theta-417059}{839755}$	$\frac{933028\theta-465533}{937026}$
j	43	44	45	46	47	48
ϕ_j	$\frac{25111}{33794}$	$\frac{81625}{109822}$	$\frac{44129}{59359}$	$\frac{31747}{42694}$	$\frac{102583}{137926}$	$\frac{110293}{148262}$
$\tau(\theta)$	$\frac{519146\theta-259055}{521256}$	$\frac{1152215\theta-575015}{1156663}$	$\frac{1275259\theta-636479}{1279941}$	$\frac{1407898\theta-702739}{1412820}$	$\frac{775309\theta-387019}{777893}$	$\frac{1703917\theta-850625}{1709337}$
j	49	50	51	52		
ϕ_j	$\frac{19730}{26517}$	$\frac{126853}{170458}$	$\frac{135721}{182342}$	$\frac{1159}{1553} \approx 0.7462$		
$\tau(\theta)$	$\frac{1868305\theta-932755}{1873983}$	$\frac{2044304\theta-1020689}{2050246}$	$\frac{1116224\theta-557347}{1119330}$			

Table 2